# Web Scraping Made Easy with Selenium

This brief tutorial introduces you to the many uses of web scraping. Also learn how Selenium, an open source web scraping tool, works.

Web scraping helps to extract large quantities of data from the internet, which can be used for various purposes. It can be useful when you want to check your competitors' prices, market trends, etc, to make the right business decisions. Web scraping is important in academic research to collect data for various economic or social studies. It is also used to collect data for training AI models — for example, to scrape tweets for training a sentiment analysis model. Other uses include monitoring volatile flight prices and checking for job opportunities from multiple websites.

As this field is evolving, it is important for us to have a basic knowledge of it. Let's begin by understanding how a popular web scraping package called Selenium works.

Selenium helps us with browser automation in multiple ways. It is compatible with almost all the programming languages including C#, Groovy, Python, Ruby, and more. It works on all major operating systems like Windows, Linux and MacOS, and can be used for testing most modern browsers — you may need to install the drivers for the associated browser. Basically, Selenium has a web driver that can work interchangeably with different web browsers. The browser drivers typically open an instance of the browser and access it.

Let us get started with the installation and use of this tool. We are going to use Selenium in Python. You can install it using the following command as shown in Figure 1.

```
!pip install selenium
```

It is easier if you use a Jupyter Notebook for this. Once the installation is done, you can install the browser driver that we discussed above. I will be using the Chrome browser driver, but you can also use Firefox or any other preferred browser. You can download the Chrome driver from *https://sites.google.com/chromium.org/driver/downloads.*

The first step is to import the installed Selenium library and then initiate the Chrome driver. This can be done using the code below:

```
from selenium import webdriver
sd=webdriver.Chrome()
```

```
In [1]: !pip install selenium

Collecting selenium
  Obtaining dependency information for selenium from https://files.pythonhosted.org/packages/a6/1e/5f1a5dd2a28528c4
b3ec6e076b58e4c035810c805328f9936123283ca14e/selenium-4.27.1-py3-none-any.whl.metadata
  Downloading selenium-4.27.1-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: urllib3[socks]<3,>=1.26 in /Users/jishnusaurav/anaconda3/lib/python3.11/site-package
s (from selenium) (1.26.16)
Collecting trio~=0.17 (from selenium)
  Obtaining dependency information for trio~=0.17 from https://files.pythonhosted.org/packages/3c/83/ec3196c360afff
bc5b342ead48d1eb7393dd74fa70bca75d33905a86f211/trio-0.27.0-py3-none-any.whl.metadata
  Downloading trio-0.27.0-py3-none-any.whl.metadata (8.6 kB)
Collecting trio-websocket~=0.9 (from selenium)
  Obtaining dependency information for trio-websocket~=0.9 from https://files.pythonhosted.org/packages/48/be/a9ae5
f50cad5b6f85bd2574c2c923730098530096e170c1ce7452394d7aa/trio_websocket-0.11.1-py3-none-any.whl.metadata
  Downloading trio_websocket-0.11.1-py3-none-any.whl.metadata (4.7 kB)
Requirement already satisfied: certifi>=2021.10.8 in /Users/jishnusaurav/anaconda3/lib/python3.11/site-packages (fr
om selenium) (2023.7.22)
Requirement already satisfied: typing_extensions~=4.9 in /Users/jishnusaurav/anaconda3/lib/python3.11/site-packages
(from selenium) (4.9.0)
Collecting websocket-client~=1.8 (from selenium)
  Obtaining dependency information for websocket-client~=1.8 from https://files.pythonhosted.org/packages/5a/84/446
87a29792a70e111c5c477230a72c4b957d88d16141199bf9acb7537a3/websocket_client-1.8.0-py3-none-any.whl.metadata
  Downloading websocket_client-1.8.0-py3-none-any.whl.metadata (8.0 kB)
Collecting attrs>=23.2.0 (from trio~=0.17->selenium)
  Obtaining dependency information for attrs>=23.2.0 from https://files.pythonhosted.org/packages/89/aa/ab0f7891a01
```

Figure 1: Selenium installation